

BAULKHAM HILLS HIGH SCHOOL

YEAR 11

HSC ASSESSMENT TASK

December 2010

**MATHEMATICS
EXTENSION 1**

Time allowed - 70 minutes

DIRECTIONS TO CANDIDATES

- Attempt ALL questions. You do NOT need to start each question on a new page.
- Allocated marks are indicated for each question.
- All necessary working should be shown.
- Approved calculators and Mathematical templates may be used
- Write your teacher's name and your name on the cover sheet provided.
- At the end of the exam, staple your answers in order behind the cover sheet provided. Staple the question sheets to the back.

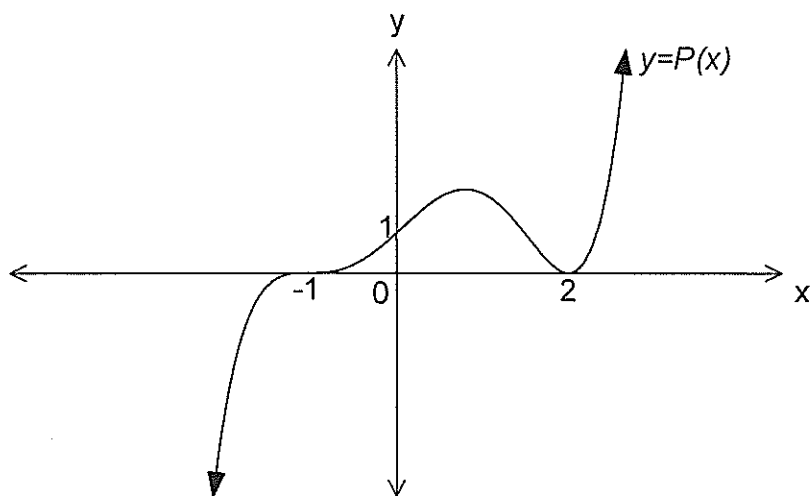
Question 1

a) Given $P(x) = x^4 - x^2 + 1$ and $Q(x) = x^2 + 1$, divide $P(x)$ by $Q(x)$. Hence express $P(x)$ in the form $Q(x) \cdot A(x) + R(x)$, where $A(x)$ and $R(x)$ are polynomials

2

b) Find a possible equation for the curve shown

3



Question 2

The polynomial $P(x) = 2x^3 - 5x^2 + 3x + 1$ has zeroes α , β and γ . Find the value of:

(i) $\alpha + \beta + \gamma$

1

(ii) $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$

2

(iii) $\alpha^2 + \beta^2 + \gamma^2$

2

Question 3

a) Find the Cartesian equation of the curve with the parametric equations:

(i) $x = 2t, y = 3t^2$

1

(ii) $x = \cos \theta, y = \sin \theta - 1$

1

b) $x^2 - 4$ is a factor of $x^4 + x^3 + px^2 + qx + 48$. Find the value of p and q .

3

Question 4

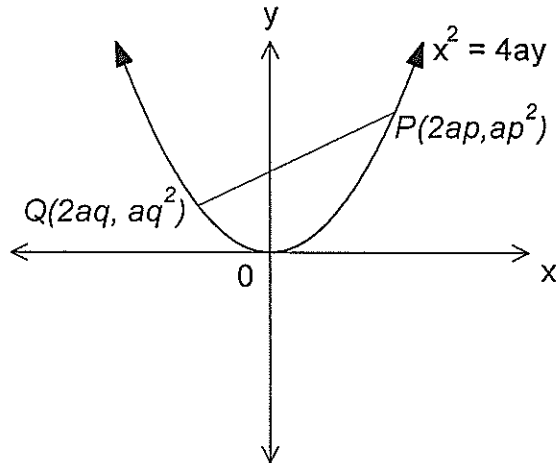
Using the principle of Mathematical Induction, prove that

$$\sum_{r=1}^n 5^r = \frac{5}{4}(5^n - 1), \text{ for all positive integers } n.$$

4

Question 5

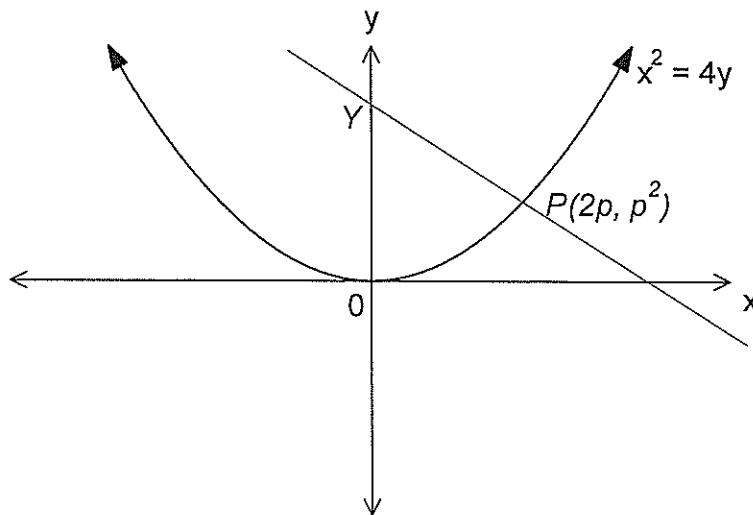
$P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$.



- | | |
|--|---|
| a) Find the equation of the chord PQ | 2 |
| b) Show that the equation of the tangent at P is $y = px - ap^2$. | 1 |
| c) The tangents at P and Q intersect at T . Find the coordinates of T . | 2 |
| d) Given that the chord PQ has gradient 2, find the equation of the locus of T , and give a geometrical description of this locus. | 2 |

Question 6

$P(2p, p^2)$ is a variable point on the parabola $x^2 = 4y$



- | | |
|--|---|
| a) Show that the equation of the normal at P is $x + py = 2p + p^3$. | 2 |
| b) The normal at P intersects the y -axis at Y . Write down the coordinates of Y . | 1 |
| c) Find the value of p if the area of $\triangle OPY$ is 33 square units, and p is a positive integer. | 2 |

Question 7

The polynomial $P(x) = Ax^3 + Bx^2 + 2Ax + C$ has real roots $\sqrt{p}, \frac{1}{\sqrt{p}}, \alpha$. Given that $C \neq 0$:

- a) Explain why $\alpha = \frac{-C}{A}$ 1
- b) Show that $A^2 + C^2 = BC$ 2

Question 8

- a) Use the method of mathematical induction to prove that if x is a positive integer then $(1+x)^n - 1$ is divisible by x for all positive integers n . 4
- b) Factorise $12^n - 4^n - 3^n + 1$. 1
- c) Hence deduce that $12^n - 4^n - 3^n + 1$ is divisible by 6 for all positive integers n . (Do NOT perform another proof by Mathematical Induction) 1

End of task

Q1.

a)

$$\begin{array}{r}
 x^2 - 2 \\
 x^2 + 1 \overline{) x^4 - x^2 + 1} \\
 \underline{x^4 + x^2} \\
 -2x^2 + 1 \\
 \underline{-2x^2 - 2} \\
 3
 \end{array}$$

correct quot. and remainder

$$P(x) = (x^2 + 1)(x^2 - 2) + 3$$

$$\begin{aligned}
 P(-2) &= 16 - 8 + 4p - 2q + 48 = 0 \\
 4p - 2q &= -56 \quad \text{--- 1} \\
 2p - q &= -28 \quad \text{--- 2} \\
 \hline
 4p &= -164 \\
 p &= -41 \\
 -32 + q &= -36 \\
 q &= -4
 \end{aligned}$$

b) $P(x) = \frac{1}{4}(x+1)^3(x-2)^2$

$\nwarrow$ $\nwarrow$ $\nwarrow$ $\nwarrow$

Q2.

(i) $\frac{5}{2}$ 1

(ii) $\frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} = \frac{3/2}{-1/2} \leftarrow 1$

$= -3$

(iii) $(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$ 1

$$= \left(\frac{5}{2}\right)^2 - 2\left(\frac{3}{2}\right)$$

$$= \frac{13}{4}$$

Q3.

a) i) $y = 3\left(\frac{x}{2}\right)^2$ 1

$y = \frac{3x^2}{4}$

ii) $\cos^2\theta + \sin^2\theta = 1$

$$x^2 + (y+1)^2 = 1$$

b) $P(2) = 16 + 8 + 4p + 2q + 48 = 0$

$$\begin{aligned}
 4p + 2q &= -72 \\
 2p + q &= -36 \quad \text{--- 1}
 \end{aligned}$$

Q4.

If $n=1$ LHS = $5^1 = 5$

$$RHS = \frac{5}{4}(5-1) = \frac{5}{4} \times 4 = 5$$

LHS = RHS $\therefore$ True for $n=1$ 1

Assume true for $n=k$

ie. Assume

$$\sum_{r=1}^k 5^r = \frac{5}{4}(5^k - 1)$$

Need to prove true for $n=k+1$

ie. Prove

$$\sum_{r=1}^{k+1} 5^r = \frac{5}{4}(5^{k+1} - 1)$$

$$LHS = 5^1 + 5^2 + 5^3 + \dots + 5^k + 5^{k+1}$$

$$= \sum_{r=1}^k 5^r + 5^{k+1}$$

$$= \frac{5}{4}(5^k - 1) + 5^{k+1}$$

by assumption

$$= \frac{1}{4} \cdot 5(5^k - 1) + 5^{k+1}$$

$$= \frac{1}{4}(5^{k+1}) - \frac{5}{4} + 5^{k+1}$$

$$= \frac{5}{4}(5^{k+1}) - \frac{5}{4}$$

$$= \frac{5}{4}(5^{k+1} - 1)$$

= RHS $\therefore$ If true for $n=k$,

then also true for $n=k+1$.

Now statement is true for $n=1$

$\therefore$ True for $n=2, 3, 4, \dots$

By induction, it is true for all positive integers n . |

Q5.

$$\begin{aligned} \text{a) } m &= \frac{ap^2 - aq^2}{2ap - 2aq} = \frac{a(p-q)(p+q)}{2a(p-q)} \\ &= \frac{p+q}{2} \quad | \end{aligned}$$

$$y - ap^2 = \left(\frac{p+q}{2}\right) \cdot (x - 2ap) \quad |$$

$$2y - 2ap^2 = (p+q)x - 2ap^2 - 2apq$$

$$2y = (p+q)x - 2apq$$

$$y = \frac{p+q}{2}x - apq$$

b) Let $q \rightarrow p$ to obtain the eqn of tangent: |

$$y \rightarrow \frac{2p}{2} \cdot x - ap^2$$

$y = px - ap^2$ is eqn of tangent

$$\begin{aligned} \text{OR } \frac{dy}{dx} &= \frac{2x}{4a} \\ &= \frac{x}{2a} \end{aligned}$$

$$= \frac{2ap}{2a} \quad \text{at } P$$

$$= p$$

$$y - ap^2 = p(x - 2ap)$$

$$\begin{aligned} y - ap^2 &= px - 2ap^2 \\ y &= px - ap^2 \end{aligned} \quad]$$

$$\begin{aligned} \text{c) } y &= px - ap^2 \\ y &= qx - aq^2 \end{aligned}$$

Intersect when

$$px - ap^2 = qx - aq^2$$

$$px - qx = ap^2 - aq^2$$

$$(p-q)x = a(p-q)(p+q)$$

$$x = a(p+q) \quad \dots \quad |$$

$$\begin{aligned} y &= p \cdot a(p+q) - ap^2 \\ &= ap^2 + apq - ap^2 \\ &= apq \quad \dots \quad | \end{aligned}$$

$$\therefore T(a(p+q), apq)$$

$$\text{d) Since } m=2, \quad \frac{p+q}{2} = 2$$

$$p+q = 4$$

$\therefore$ At T : $x = 4a$ which is constant.

$\therefore$ Locus is $x = 4a$ |

But tangents can only intersect outside the parabola.

$\therefore$ Locus is the ray $x = 4a$, lying below the parabola, (and parallel to the y -axis) not needed.

["Ray" or "part of line" lying below or outside the parabola]

Q6.

$$a) y = \frac{x^2}{4} \quad \frac{dy}{dx} = \frac{2x}{4} = \frac{2(2p)}{4} \text{ at } P. = p$$

Normal:

$$m = -\frac{1}{p}$$

$$y - p^2 = -\frac{1}{p}(x - 2p)$$

$$py - p^3 = -x + 2p$$

$$x + py = p^3 + 2p$$

b) If $x=0$, $py = p^3 + 2p$

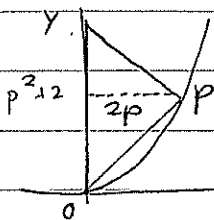
$$y = p^2 + 2$$

$\therefore Y(0, p^2 + 2)$

c) Area

$$= \frac{1}{2} \cdot (p^2 + 2) \cdot 2p$$

$$= p^3 + 2p$$



If $p^3 + 2p = 33$, $p^3 + 2p - 33 = 0$

($p = 1, 3, 11, 33$ - possible values)

$$3^3 + 2(3) - 33 = 0$$

$\therefore p = 3$

Q7.

a) Product of roots $= \sqrt{p} \cdot \frac{1}{\sqrt{p}} \cdot \alpha = \frac{-c}{A}$

$\therefore \alpha = -\frac{c}{A}$

b) α is a root

$$A\left(-\frac{c}{A}\right)^3 + B\left(-\frac{c}{A}\right)^2 + 2A\left(-\frac{c}{A}\right) + C = 0$$

$$\frac{-C^3}{A^2} + \frac{BC^2}{A^2} - 2C + C = 0$$

$$\frac{-C^3}{A^2} + \frac{BC^2}{A^2} = C$$

$$-C^3 + BC^2 = A^2C$$

$$\div C \neq 0$$

$$-C^2 + BC = A^2$$

$$\text{i.e. } A^2 + C^2 = BC$$

Q8.

a) If $n=1$

$$(1+x)^1 - 1 = 1+x-1 = x$$

which is divisible by x

$\therefore$ True for $n=1$

Assume true for $n=k$:

$$(1+x)^k - 1 = mx \quad (m = \text{integer})$$

Need to prove true for $n=k+1$

i.e. Prove $(1+x)^{k+1} - 1 = px$

($p = \text{integer}$)

$$\begin{aligned} \text{LHS} &= (1+x)(1+x)^k - 1 \\ &= (1+x)^k + x(1+x)^k - 1 \\ &= [(1+x)^k - 1] + x(1+x)^k \\ &= mx + x(1+x)^k \\ &= x(m + (1+x)^k) \quad \leftarrow \text{by assumption} \\ &= x(\text{integer}) = px \end{aligned}$$

$\therefore$ If true for $n=k$, then true for $n=k+1$

Conclusion (no mark)

b) $4^n(3^n - 1) - (3^n - 1)$

$$= (4^n - 1)(3^n - 1)$$

c) $4^n - 1 = (1+3)^n - 1 \therefore \text{Div by } 3$

$3^n - 1 = (1+2)^n - 1 \therefore \text{Div by } 2$

$\therefore$ Product is div by $2 \times 3 = 6$